

Western Ghats Field Validator

Mobile Application for Ground-Truth Validation of Satellite-Derived Land Cover

Key Capabilities: Offline-first architecture | CoRE Stack API integration (8 endpoints) | Forest vs plantation classification | GPS-based observation capture | JSON/CSV export

1. Overview and Interface

The Western Ghats Field Validator is a mobile-first progressive web application that enables field researchers to validate satellite-derived LULC classifications through systematic ground-truth observations. It integrates CoRE Stack watershed APIs, Google Dynamic World, and custom forest typology datasets.

Interface Features: Interactive map with street/satellite base layers | GPS location tracking | Layer toggle with active count | Bottom navigation (Map, Layers, Capture, Guide, Log)

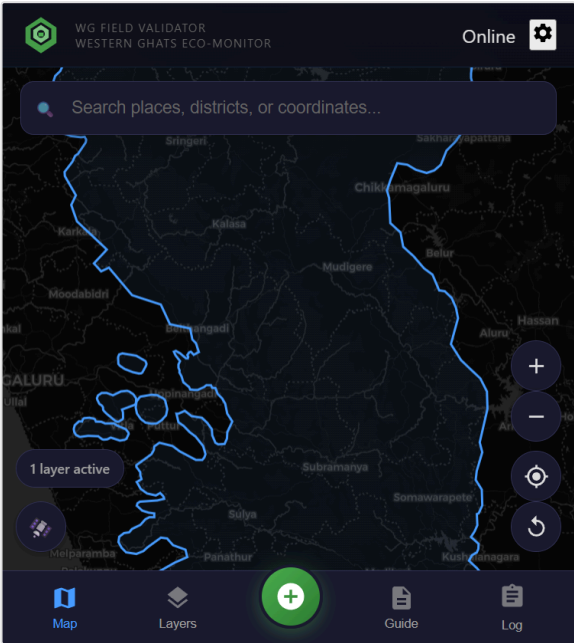


Figure 1: Main interface

2. Layer Management

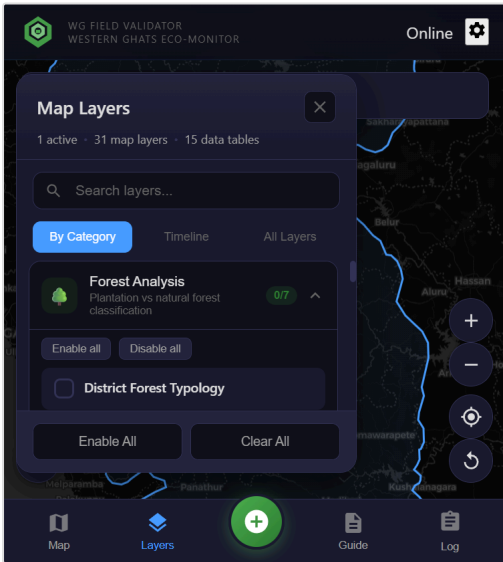


Figure 2: Layer categories

The application organizes **31 map layers** and **15 data tables**:

Category	Description	Count
Forest Analysis	Plantation vs natural forest classification	7
Land Cover Maps	Historical LULC from GLC-FCS30D (1985-2022)	16
Urban Expansion	Built-up area change detection	14
Dynamic World	Google Earth Engine LULC classification	2
Boundaries	Administrative boundaries (District, WG)	2
Watershed Data	Water balance, cropping intensity (CoRE Stack)	4

3. Forest vs Plantation Classification

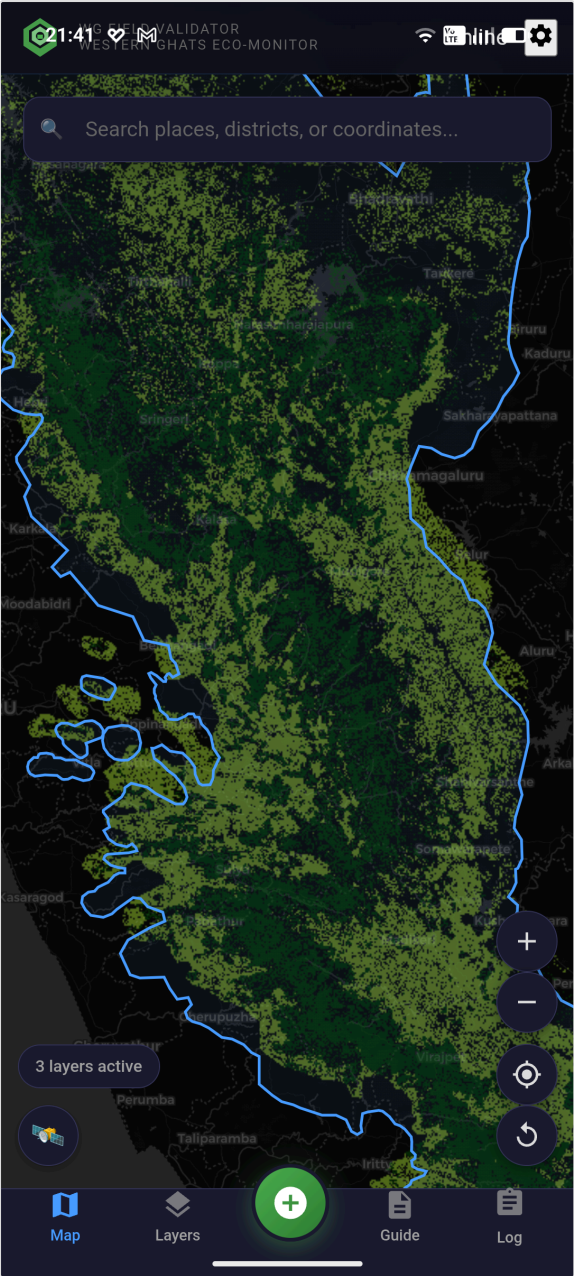


Figure 3: Natural forest (green) vs plantations (purple)

The forest typology distinguishes natural forests from plantations using multi-criteria analysis:

Natural Forest Indicators

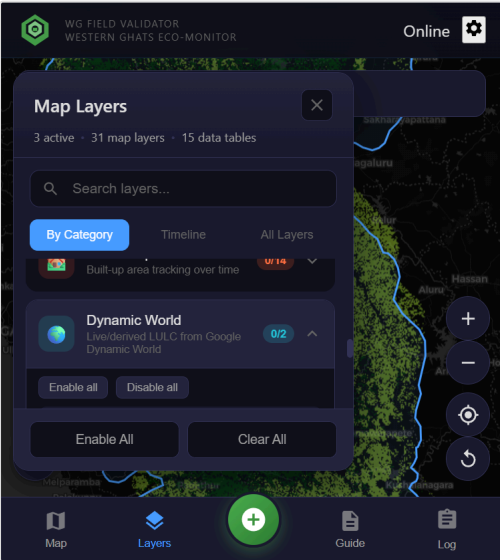
- **Tree cover persistence:** Consistent 2000-2020
- **Canopy structure:** Multi-layered, heterogeneous
- **Spatial pattern:** Irregular boundaries
- **Historical use:** No clearing/replanting

Plantation Indicators

- **Spectral signatures:** Monoculture reflectance
- **Row patterns:** Geometric arrangements
- **Age uniformity:** Even-aged stands
- **Crop cycles:** Periodic harvesting

This distinction is critical: natural forests support significantly higher biodiversity than plantations, despite both appearing as "forest" in conventional satellite classifications.

4. Dynamic World and CoRE Stack Integration



Dynamic World LULC

Google's near real-time land cover with 9 classes: Water, Trees, Grass, Flooded Vegetation, Crops, Shrub/Scrub, Built-up, Bare Ground, Snow/Ice. Available as Live GEE (real-time) or Regional composite (2018-2025).

CoRE Stack API Endpoints

Endpoint	Data Provided
get_admin_details_by_latlon	State, District, Tehsil lookup
get_mwsid_by_latlon	Micro-Watershed ID
get_generated_layer_urls	GIS layers for tehsil
get_tehsil_data	Tehsil-level statistics
get_mws_data	ET, Runoff, Precipitation time series
get_mws_kyl_indicators	Watershed indicators
get_waterbodies_data_by_admin	Waterbody inventory
get_mws_report	MWS report URL

5. Field Observation Workflow

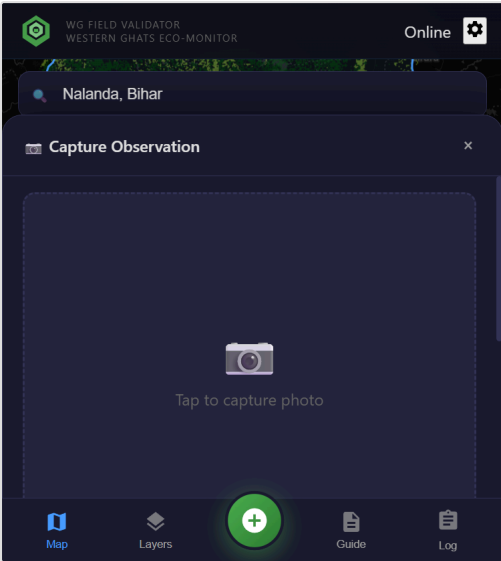


Figure 5: Capture interface

Capture Process

- 1. **Photo capture:** Camera or gallery selection
- 2. **GPS coordinates:** Automatic with accuracy indicator
- 3. **Layer values:** Active layer data at location displayed
- 4. **Validation status:** Match (agrees) | Mismatch (differs) | Unclear
- 5. **Field notes:** Optional text documentation

Observations are stored locally in IndexedDB and sync when connectivity is restored.

6. Field Log and Data Export

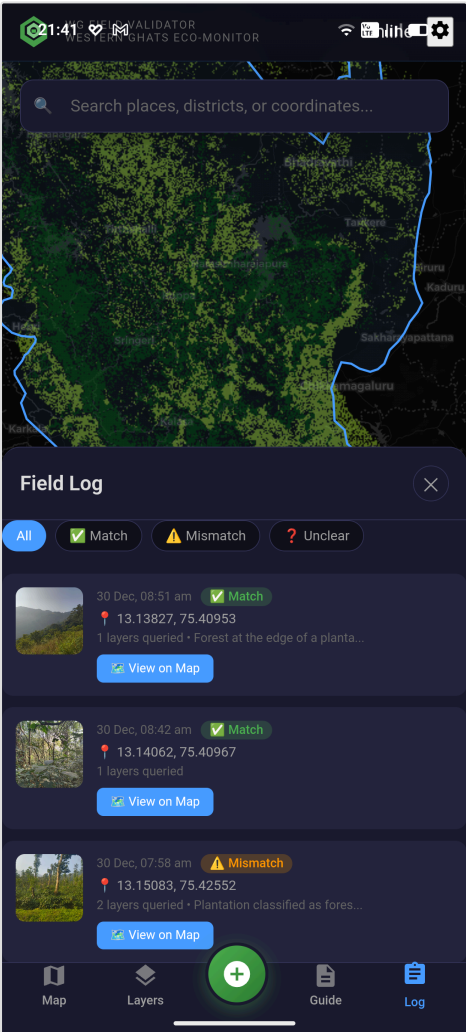


Figure 6: Field log with filtering

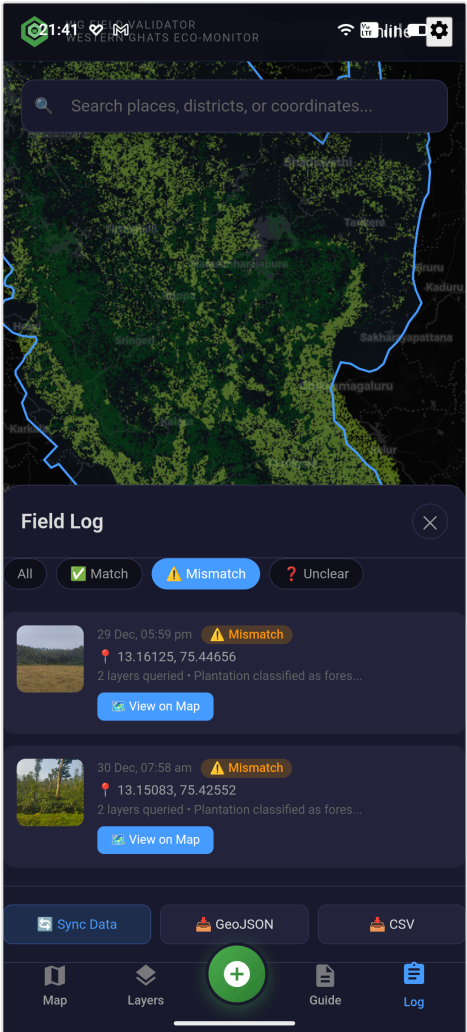


Figure 7: Export options

Log Features

- Chronological list with thumbnails
- Filter by status (All/Match/Mismatch/Unclear)
- View details and map location

Export Formats

- **JSON:** Full metadata for analysis
- **CSV:** Spreadsheet/GIS compatible

7. Additional Features and Technical Architecture

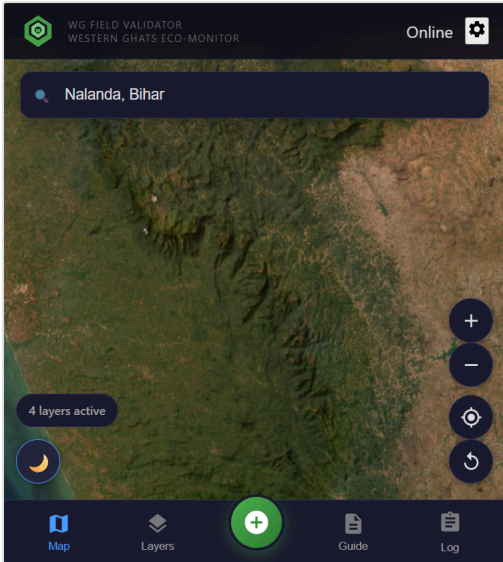


Figure 8: Satellite base map

Base Maps

- OpenStreetMap: Navigation
- ESRI Imagery: Verification
- Dark mode: Low-light

Field Guide

- Getting started / GPS setup
- Layer management
- Offline operation

Technical Stack

Frontend	React 18 + TypeScript + Vite
Mapping	MapLibre GL JS
Mobile	Capacitor (Android)
Storage	IndexedDB
APIs	CoRE Stack, Nominatim

8. Benefits and Resources

For Field Researchers

- Works offline in remote areas
- Automatic GPS with accuracy
- Multiple layer comparison

For Conservation Orgs

- Systematic validation
- Plantation vs forest distinction
- Standardized methodology

Resources

- Sample data: `field-data/recovered-observations/`
- API notebook: `docs/notebooks/corestack_api_tested.ipynb`
- License: MIT